Jialing Wu

Ph.D. Student, Engineering Education
The Ohio State University

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Education

The Ohio State University, Columbus, OH — Expected May 2028
Ph.D., Engineering Education
Minor in Computer Science and Engineering (AI Track)
Advisor: Adam R. Carberry

Vanderbilt University, Peabody College, Nashville, TN — May 2024
M.Ed., International Education Policy and Management

Shanghai University, Shanghai, China — July 2017
B.E., Mechanical Engineering

Research Areas

I study how emerging technologies such as AI and robotics are reshaping engineering and STEM education, from classroom teaching to education policy, with a focus on equity and access. My work combines qualitative and critical quantitative methods with AI-assisted analysis.

Publications

Peer-Reviewed Journal Articles

[22] **Wu, J.**, Shehab, F., & McGuire, L. (2026). Too deep or too shallow: Reflections on methods training from the perspective of engineering education students. *Teaching Educational Research Methods*, *1*(2), Article 5040. https://doi.org/10.5149/term.5040

[21] Yan, J., Wei, X., & **Wu, J.** (2026). Curriculum- and grade-level AI policy implementation: A comparative analysis of North Carolina and California. *Educational Policy*. Advance online publication. https://doi.org/10.1177/08959048261481543

[20] **Wu, J.**, Dalal, M., Figard, R., & Carberry, A. R. (2026). Understanding transgender and gender-nonconforming students' experiences in precollege engineering education: A latent profile analysis [Invited contribution based on ASEE PCEE Division Best Paper award]. *Journal of Pre-College Engineering Education Research (J-PEER)*, *16*(1), Article 3. https://doi.org/10.7771/2157-9288.1515

Editorial

[19] Darling-Aduana, J. S., Heinrich, C., Noonan, J., **Wu, J.**, & Enriquez, K. (2026). Failing to learn from failure in online credit recovery assessments [Policy Insights]. *Journal of Policy Analysis and Management*, *45*(3), e70120. https://doi.org/10.1002/pam.70120

Book Chapters

[18] Yan, J., **Wu, J.**, & Wei, X. (2026). Silent policies: How GenAI guidelines in U.S. higher education overlook international students' experiences. In A. Chen (Ed.), *Generative AI for supporting international student well-being in higher education* (pp. 95–132). IGI Global. https://doi.org/10.4018/979-8-3373-6571-8.ch004

Peer-Reviewed Conference Papers

[17] **Wu, J.**, Ren, S., & Wang, S. (2026). Using bipartite network analysis to examine global collaboration between engineering and STEM education (2014–2024). *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--61184

[16] **Wu, J.**, & Wang, S. (2026). What does the "person-centered quantitative approach" mean in practice? A literature review in STEM education research [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--60763

[15] **Wu, J.**, & Carberry, A. R. (2026). Integrating robotics and engineering education: Initial validation testing of an SCCT- and CT-based survey instrument [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--60118

[14] **Wu, J.**, & Melo de Lyra, M. V. (2026). Exploring perspectives on pre-college robotics education: A social media–based topic and emotion analysis [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--59786

[13] Melo de Lyra, M. V., **Wu, J.**, & Carberry, A. R. (2025). Exploring early-career engineering faculty experiences teaching using social cognitive theory. *2025 IEEE Frontiers in Education Conference (FIE)*, Nashville, TN, pp. 1–10. https://doi.org/10.1109/FIE63693.2025.11328485

[12] Dalal, M., **Wu, J.**, & Iqbal, A. (2025). Overcoming challenges in robotics education: Examining teacher facilitators and barriers. *Research in Engineering Education Symposium (REES)*, Johor Bahru, Malaysia.

[11] **Wu, J.**, Dalal, M., & Carberry, A. R. (2025). Advancing equity: Exploring the experiences of transgender and gender non-conforming students in a pre-college engineering course [Work-in-progress paper]. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--57577

[10] Wei, X., **Wu, J.**, & Klein-Gardner, S. S. (2025). Parents' knowledge, attitude, and behavior on pre-college engineering education course [Work-in-progress paper]. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--55568

[9] Lachapelle, C. P., Dalal, M., McKeown, K., & **Wu, J.** (2025). Curriculum design for all learners. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--56190

[8] **Wu, J.**, & Dalal, M. (2024). High school students' perspectives on pre-college engineering education courses (Fundamental). *2024 ASEE Annual Conference & Exposition Proceedings*, Portland, OR. https://doi.org/10.18260/1-2--47527

[7] **Wu, J.**, Léger, N., & Klein-Gardner, S. S. (2024). High school students' perspectives on mathematical modeling in the engineering design process (RTP). *2024 ASEE Annual Conference & Exposition Proceedings*, Portland, OR. https://doi.org/10.18260/1-2--47528

Conference Presentations

[6] **Wu, J.**, & Li, D. (2026, November 5–7). Teachers as AI policy actors: Focus group evidence from a K–12 robotics consortium [Poster presentation]. *Association for Public Policy Analysis & Management (APPAM) Annual Fall Research Conference*, Boston, MA.

[5] Wang, J., **Wu, J.**, & Cheng, Y. (2026, July 28–31). Cross-cultural evaluation of LLMs: A mixed-method study of Chinese-to-English movie title translation [Poster presentation]. *12th International Conference for Computational Social Science (IC2S2 2026)*, Burlington, VT.

[4] Wei, X., & **Wu, J.** (2026, April 8–12). Navigating AI in higher education: Student views on professors' attitudes toward AI tools in the classroom analyzed through sentiment analysis and topic modeling [Conference presentation]. *American Educational Research Association (AERA) Annual Meeting*, Los Angeles, CA.

[3] **Wu, J.**, & Yan, J. (2026, March 28 – April 1). Comparing high school accessibility in California and New York: A GIS-based assessment [Conference presentation]. *Comparative and International Education Society (CIES) Annual Conference*, San Francisco, CA.

[2] Wei, X., **Wu, J.**, Klein-Gardner, S. S., & Dalal, M. (2025, November 10–14). Evaluating fidelity of teaching in pre-college engineering: A case study of curriculum implementation [Poster presentation]. *Evaluation 2025*, Kansas City, MO. eval25.eventscribe.net

[1] Darling-Aduana, J. S., Heinrich, C., Noonan, J., **Wu, J.**, & Enriquez, K. (2025, April 23–27). Failing to learn from failure: The facade of online credit recovery assessments [Conference presentation]. *American Educational Research Association (AERA) Annual Meeting*, Denver, CO. https://doi.org/10.3102/ip.25.2182630

Manuscripts in Revision

[2] Berhane, B., Dalal, M., Ross, L., **Wu, J.**, & Wang, J. (revise and resubmit). Examining student- and school-level outcomes in pre- and post-survey data from a pre-college engineering program. *Journal of Engineering Education*.

[1] **Wu, J.**, Wei, X., & Yan, J. (revising for resubmission). Engineering education deserts in the United States: A geographic random forest analysis of broad-access engineering programs in higher education. *AERA Open*.

Research Experience

Graduate Research Associate August 2024 – Present
Department of Engineering Education, The Ohio State University, Columbus, OH
e4usa+FIRST (NSF-funded); Supervisor: Adam R. Carberry

- Lead the development and validation of a survey instrument grounded in Social Cognitive Career Theory and computational thinking, measuring students' self-efficacy, identity, and skills.
- Coordinate research approvals and IRB protocols across multiple school districts for multi-site data collection.
- Analyze teacher interviews and focus groups on robotics curriculum implementation, identifying facilitators and barriers.

Lead Researcher, Independent and Collaborative Research 2024 – Present
The Ohio State University, Columbus, OH
Computational social science in education

- Map educational access using GIS and geospatial machine learning, including engineering education deserts and high school accessibility.
- Analyze research collaboration in STEM education and public discourse on pre-college robotics using network and text analysis.
- Examine person-centered quantitative approaches in STEM education and graduate methods training.
- Collaborate on AI-related studies, including AI policy implementation in K–12 and higher education and LLM-assisted qualitative coding.

Graduate Research Assistant June 2023 – July 2024
School of Engineering, Vanderbilt University, Nashville, TN
e4usa (NSF-funded); Supervisors: Stacy S. Klein-Gardner and Medha Dalal

- Developed codebooks and analyzed student surveys and focus groups, including pre/post comparisons across cohorts; first-authored two ASEE 2024 conference papers.

Graduate Research Assistant September 2023 – May 2024
Peabody College, Vanderbilt University, Nashville, TN
Supervisors: Carolyn J. Heinrich and Jennifer S. Darling-Aduana

- Coded 1,400+ online credit recovery exam questions and tested how readily answers could be found online, contributing to an AERA 2025 presentation and a *JPAM* article.

Teaching Experience

Graduate Teaching Associate, The Ohio State University August 2026 – Present
ENGR 1181: Fundamentals of Engineering I

- Teach four weekly lab sections of about 30 first-year engineering students each.
- Guide hands-on labs on measurement, data analysis, design, teamwork, and technical communication, including beam bending, spot speed, smart light, and humanitarian relief labs.
- Grade lab assignments, hold office hours, and coordinate with faculty instructors and undergraduate teaching assistants.

Workshops 2026

- *From Reddit to Policy Insights: Using AI to Summarize Public Opinion on Robotics Courses.* Facilitator, AI Peer Learning Lab, APPAM Fall Research Conference, Boston, MA (November 2026).
- *Multilevel Modeling.* Organizer and presenter, OTTER (Open Toolkit for Training in Education Research), The Ohio State University (October 2026).

Curriculum Development 2023 – 2025

- *e4usa Unit 2B: Engineering is Creative — Wind Energy.* Revised a pre-college engineering unit from water filters to wind turbines, redesigning lessons and MATLAB code; published on TeachEngineering (13 lessons, 22 activities; 2024).
- *XRP Robotics Supplementary Curriculum Kit.* Designed scaffolded programming activities for the e4usa+FIRST robotics platform, moving from pseudocode to block-based coding to Python (2025).

Professional Experience

Assessment Intern May 2024 – August 2024
BioInteractive, Howard Hughes Medical Institute (HHMI), Chevy Chase, MD

- Developed an HTML coding guide (9 chapters, 25 sections) for the content team's Canvas course pages.
- Audited course accessibility with UDOIT and wrote an accessibility handbook for content editors.

Consultant July 2017 – July 2022
Shanton Way Executive Search, Shanghai, China

- Recruited executives for clients in international K–12 education, consulting, finance, and technology; built and managed a pool of 1,500+ bilingual candidates for international schools.

Training and Certificates

College and University Teaching Graduate Certificate, The Ohio State University In Progress

Summer Institute in Computational Social Science (SICSS) 2025
University of Rochester: LLMs and network analysis

ICPSR Summer Program in Quantitative Methods of Social Research 2024
University of Michigan: text analysis, Python, and R

Fellowships, Awards, and Grants

Fellowships and Awards

Frédéric Bastiat Fellowship ($5,000), Mercatus Center at George Mason University 2025–2026

Best Paper Award, 2nd Place, Pre-College Engineering Education Division, ASEE 2025

Scholastic Achievement Scholarship ($26,500), Vanderbilt University, Peabody College 2022–2024

University Scholarship for Academic Excellence, Shanghai University 2014, 2015

Travel and Professional Development Grants

Sponsored Graduate Student Registration, AERA Division D 2026

Smith Travel Grant ($1,000 each), Department of Engineering Education, OSU 2025, 2026

Career Development Grant ($350), Council of Graduate Students, OSU 2025

Service and Leadership

Professional Service

Reviewer, Journal of Engineering Education 2026

Reviewer, AERA Annual Meeting 2026

Reviewer, ASEE Annual Conference & Exposition 2025, 2026

Reviewer, SACNAS NDiSTEM Conference 2025

Discussant, APPAM Annual Fall Research Conference 2026

Student Session Moderator, ASEE Annual Conference & Exposition 2026

University and Community Service

Institutional Coordinator, Engineering Education Graduate Student Exchange Program, OSU 2025–Present

Graduate Ambassador, Department of Engineering Education, OSU 2025–Present

Graduate Student Host, Faculty Candidate Visits, Department of Engineering Education, OSU 2024–2026

Volunteer, Florence Nightingale Day for Statistics and Data Science, OSU 2025

Founding Member and Treasurer, International Mentorship at Peabody, Vanderbilt University 2023–2024

Mentor, Peabody Peer Career Mentor Program, Vanderbilt University 2023–2024

Mentor, Peabody International Students and Affairs, Vanderbilt University 2023–2024

Coordinator and Teacher, Migrant Children Program, AIQUZHI Volunteer Group, Shanghai 2017–2022

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